Modular Programming – Practical Work
Week 3
Classes and Objects

Notations: W1Q1 = Week 1, Questions of type 1

Question types:

• Type 1: theory, concepts	*knowledge*
• Type 2: understanding typically based on code reading	*understanding*
• Type 3: producing yourself code	*know-how*

A moodle deposit space is open every week for you to upload your solutions. Solutions of the exercises are provided one week later.

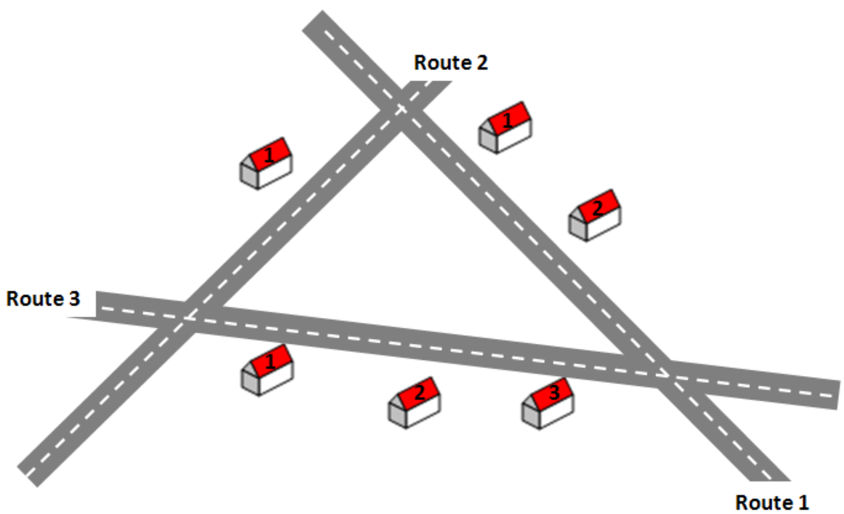
Important points:

- 602_F and 602_D classes should upload solutions **before Monday 24h00 the following week**
- **2 or 3 practical work will be corrected** and will count for one quarter of the intermediate test note

W3Q3 – 1

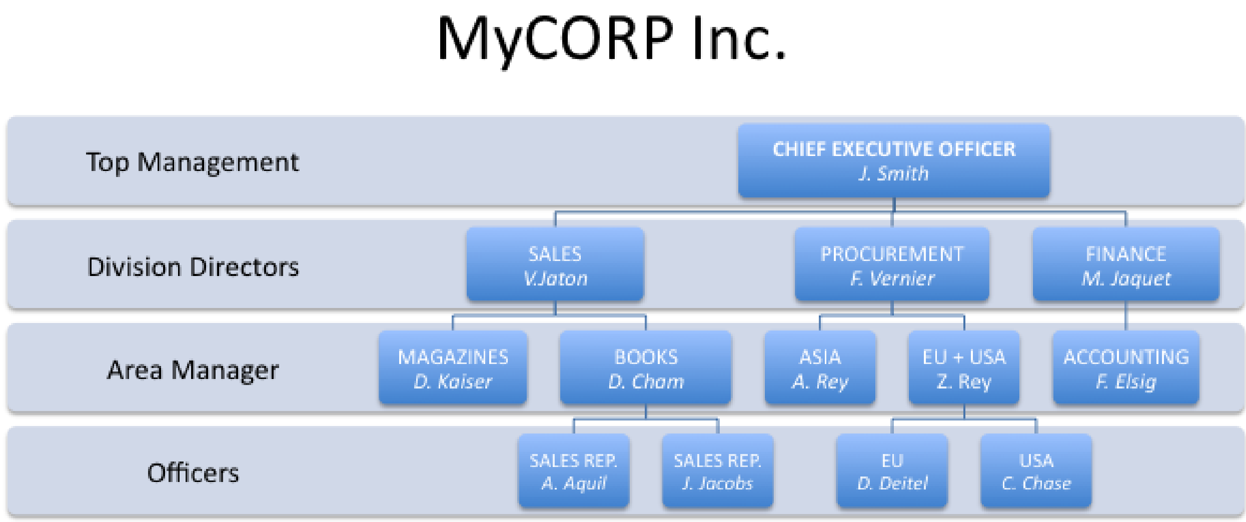
1. Create a set of classes to represent a city with the following constraints:
   - A city has a *name* and is composed of a set of *streets*
   - A street is composed of a set of *houses*
   - Each house has
     - a given number of inhabitants
     - a unique number in the street
     - a family name
   - A street is identified by a street name which is unique
   - A street is connected to two other streets

2. Implements a test class that creates a city according to the following illustration.

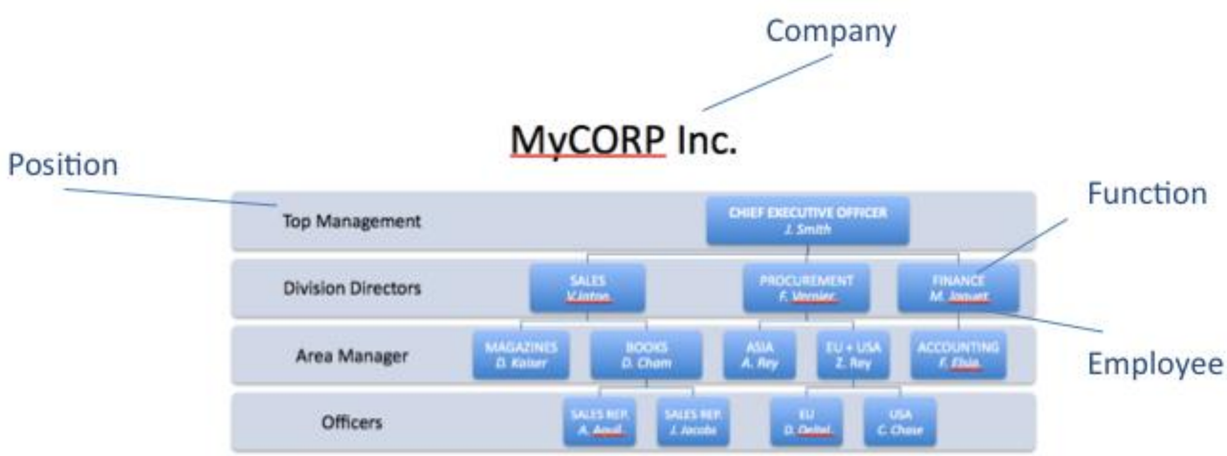


W3Q3 – 2

Create a set of classes to represent a company organization chart (organigram) as illustrated below.



To help your design, make the assumption that the following entities have been identified:



Furthermore, the following relationships are imposed:

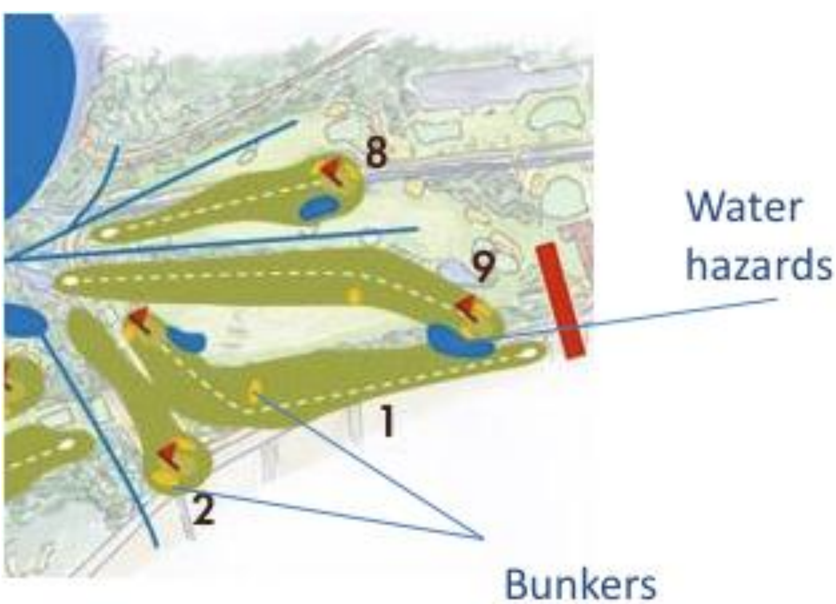
- A *Company* has one top level *Function* (in this case, the Chief Executive Officer)
- A *Function* has a relationship of responsibility to other *Functions* (for example, function Sales is responsible of Magazines and Books); conversely, a *Function* is dependent to another *Function* (Magazines depends to Sales)
- *Functions* are grouped into *Positions* (for example the Division Directors position has the Sales, Procurement and Finance functions)
- Write a method `displayDependency(Function f)` that outputs the dependency of a function:
  ```
  The employee Vincent Jaton depends from John Smith
  ```
- Write a method `displayResponsibilities(Function f)` that outputs the responsabilities of a function:
  ```
  The employe John Smith is responsible for
  Vincent Jaton
  Fabian Vernier
  Marcel Jaquet
  ```
- For all classes, write a `toString()` method.

W3Q3 – 3

The company employing you receives as mandate to redo the website of a golf club: http://www.golfsierre.ch/fr/. Your boss believes that similar type of golf club clients will mandate the company to build their websites. Therefore, he opts for a clean implementation that could be re-used for other clients. The boss asks you to propose a basis class design that represents what a golf is and he asks you to do the implementation in Java technology[1].

As you can see during your visit, a golf ("parcours de golf") has a given number of holes, typically 9 or 18. Looking at hole descriptions in the web site, you identify different elements that define the entity "hole" (http://www.golfsierre.ch/fr/parcours/golf1.php) : hole description, handicap, men distance, ladies distance, par (number of hits you are supposed to accomplish), etc.

A golf player explains you that a hole is also identified by hazards such as bunkers (sand traps) or water hazards. A water hazard is typically shared between different holes, as illustrated below.



1. Propose a class design.
   Your objective in doing so is to be able to recompose all the information available on the hole description web pages of the golf of Sierre.

[1] Someone else in the company will be in charge of the html page generation on top of your class (for ex using JSP or simple Java servlets).